

Tebogo Masilo

Practical 1 : Basic SQL Syntax

Question 1

My Workspace > Practical.sql

ACCOUNTADMIN COMPUTE_WH (X-Small) Choose database ...

```
1
2
3  --Q1. Display all columns for all transactions.
4  --Expected output: All columns
5  SELECT *
6  FROM practical1.DATASET.RETAIL_SALES;
```

Results (5 minutes ago)

Table Chart 1,000 rows 20ms

#	TRANSACTION_ID	DATE	CUSTOMER_ID	GENDER	AGE	PRODUCT_CATEGORY	QUANTITY	PRICE_PER_UNIT
1	1	2023-11-24	CUST001	Male	34	Beauty	3	50
2	2	2023-02-27	CUST002	Female	26	Clothing	2	500
3	3	2023-01-13	CUST003	Male	50	Electronics	1	30
4	4	2023-05-21	CUST004	Male	37	Clothing	1	500
5	5	2023-05-06	CUST005	Male	30	Beauty	2	50
6	6	2023-04-25	CUST006	Female	45	Beauty	1	30

Question 2

Home Practical.sql +

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ACCOUNTADMIN COMPUTE_WH (X-Small) Choose database ...

```
7
8
9  --Q2. Display only the Transaction ID, Date, and Customer ID for all records.
10 --Expected output: Transaction ID, Date, Customer ID
11 SELECT TRANSACTION_ID,
12         DATE,
13         CUSTOMER_ID
14 FROM PRACTICAL1.DATASET.RETAIL_SALES;
15 Ctrl+I to generate
16
```

Results (just now)

Table Chart 1,000 rows 430ms

#	TRANSACTION_ID	DATE	CUSTOMER_ID
1	1	2023-11-24	CUST001
2	2	2023-02-27	CUST002
3	3	2023-01-13	CUST003
4	4	2023-05-21	CUST004
5	5	2023-05-06	CUST005
6	6	2023-04-25	CUST006
7	7	2023-03-13	CUST007

Question 3

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ACCOUNTADMIN • COMPUTE_WH (X-Small) Choose database ...

```
16 -----
17 ---Q3. Display all the distinct product categories in the dataset.
18 --Expected output: Product Category
19 SELECT DISTINCT PRODUCT_CATEGORY
20 FROM PRACTICAL1.DATASET.RETAIL_SALES;
21 Ctrl+I to generate
```

Results (just now)

Table Chart 3 rows 86ms

	PRODUCT_CATEGORY
1	Clothing
2	Beauty
3	Electronics

Question 4

My Workspace > Practical.sql

ACCOUNTADMIN • COMPUTE_WH (X-Small) Choose database ...

```
21 -----
22 ---Q4. Display all the distinct gender values in the dataset.
23 --Expected output: Gender
24
25 SELECT DISTINCT GENDER
26 FROM PRACTICAL1.DATASET.RETAIL_SALES;
27
28
29
```

Results (just now)

Table Chart 2 rows 74ms

	GENDER
1	Male
2	Female

Question 5

```
--Q5. Display all transactions where the Age is greater than 40.
--Expected output: All columns
SELECT*
FROM PRACTICAL1.DATASET.RETAIL_SALES
WHERE AGE>40;
```

Results (10 hours ago)

Table | Chart | 534 rows | 69ms

	# TRANSACTION_ID	DATE	A CUSTOMER_ID	A GENDER	# AGE	A PRODUCT_CATEGORY	# QUANTITY
	3	2023-01-13	CUST003	Male	50	Electronics	1
	6	2023-04-25	CUST006	Female	45	Beauty	1
	7	2023-03-13	CUST007	Male	46	Clothing	2
	9	2023-12-13	CUST009	Male	63	Electronics	2
	10	2023-10-07	CUST010	Female	52	Clothing	4
	14	2023-01-17	CUST014	Male	64	Clothing	4

Question 6

```
34
35 -----
36 --Q6. Display all transactions where the Price per Unit is between 100 and 500.
37 --Expected output: All columns
38 SELECT*
39 FROM PRACTICAL1.DATASET.RETAIL_SALES
40 WHERE price_per_unit BETWEEN 100 AND 500;
41
```

Results (1 minute ago)

Table | Chart | 396 rows | 1.3s

	A CUSTOMER_ID	A GENDER	# AGE	A PRODUCT_CATEGORY	# QUANTITY	# PRICE_PER_UNIT	# TOTAL_A
1	CUST002	Female	26	Clothing	2	500	
2	CUST004	Male	37	Clothing	1	500	
3	CUST009	Male	63	Electronics	2	300	
4	CUST013	Male	22	Electronics	3	500	
5	CUST015	Female	42	Electronics	4	500	
6	CUST016	Male	19	Clothing	3	500	

Question 7

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```
--Q7. Display all transactions where the Product Category is either 'Beauty' or
42 --'Electronics'.
43 --Expected output: All columns
44
45 SELECT*
46 FROM PRACTICAL1.DATASET.RETAIL_SALES
47 WHERE PRODUCT_CATEGORY IN('Beauty','Electronics');
48 Ctrl+I to generate
```

Results (just now)

Table Chart 649 rows 25ms

#	TRANSACTION_ID	DATE	CUSTOMER_ID	GENDER	AGE	PRODUCT_CATEGORY	QUANTITY
1	1	2023-11-24	CUST001	Male	34	Beauty	3
2	3	2023-01-13	CUST003	Male	50	Electronics	1
3	5	2023-05-06	CUST005	Male	30	Beauty	2
4	6	2023-04-25	CUST006	Female	45	Beauty	1
5	8	2023-02-22	CUST008	Male	30	Electronics	4
6	9	2023-12-13	CUST009	Male	63	Electronics	2

Question 8

```
40
49
50 -----
51 ---Q8. Display all transactions where the Product Category is not 'Clothing'.
52 ---Expected output: All columns
53
54 SELECT*
55 FROM PRACTICAL1.DATASET.RETAIL_SALES
56 WHERE PRODUCT_CATEGORY NOT IN('Clothing');
```

Results (just now)

Table Chart 649 rows 21ms

#	TRANSACTION_ID	DATE	CUSTOMER_ID	GENDER	AGE	PRODUCT_CATEGORY	QUANTITY
1	1	2023-11-24	CUST001	Male	34	Beauty	3
2	3	2023-01-13	CUST003	Male	50	Electronics	1
3	5	2023-05-06	CUST005	Male	30	Beauty	2
4	6	2023-04-25	CUST006	Female	45	Beauty	1
5	8	2023-02-22	CUST008	Male	30	Electronics	4
6	9	2023-12-13	CUST009	Male	63	Electronics	2

Question 9

```

55
56 -----
57 --Q9. Display all transactions where the Quantity is greater than or equal to 3.
58 ---Expected output: All columns
59 SELECT*
60 FROM PRACTICAL1.DATASET.RETAIL_SALES
61 WHERE QUANTITY >=3;
    | Ctrl+I to generate

```

Results (just now)

Table Chart 504 rows 24ms

#	TRANSACTION_ID	DATE	CUSTOMER_ID	GENDER	AGE	PRODUCT_CATEGORY	QUANTITY
1	1	2023-11-24	CUST001	Male	34	Beauty	3
2	8	2023-02-22	CUST008	Male	30	Electronics	4
3	10	2023-10-07	CUST010	Female	52	Clothing	4
4	12	2023-10-30	CUST012	Male	35	Beauty	3
5	13	2023-08-05	CUST013	Male	22	Electronics	3
6	14	2023-01-17	CUST014	Male	64	Clothing	4

Question 10

```

61
62 -----
63 ---Q10. Count the total number of transactions.
64 ---Expected output: Total_Transactions
65 SELECT COUNT(TRANSACTION_ID) AS Total_Transactions
66 FROM PRACTICAL1.DATASET.RETAIL_SALES;

```

Results (just now)

Table Chart 1 row 35ms

#	TOTAL_TRANSCATIONS
1	1000

Question 11

```
67  --Q11. Find the average Age of customers.
68  ---Expected output: Average_Age
69  SELECT AVG(AGE) AS Average_Age
70  FROM PRACTICAL1.DATASET.RETAIL_SALES;
    Ctrl+I to generate
```

Results (just now)

Table Chart 1 row 72ms

	# AVERAGE_AGE
1	41.392000

Question 12

```
71  -----
72  --Q12. Find the total quantity of products sold. Expected output: Total_Quantity
73  SELECT SUM(QUANTITY) AS Total_Quantity
74  FROM PRACTICAL1.DATASET.RETAIL_SALES;
    Ctrl+I to generate
```

Results (just now)

Table Chart 1 row 73ms

	# TOTAL_QUANTITY
1	2514

Question 13

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```
74
75 -----
76 ---Q13. Find the maximum Total Amount spent in a single transaction. Expected output: Max_Total_Amount
77 SELECT MAX(TOTAL_AMOUNT) AS Max_Total_Amount
78 FROM PRACTICAL1.DATASET.RETAIL_SALES;
79 | Ctrl+I to generate
```

Results (1 hour ago)

Table Chart 1 row 31ms

	# MAX_TOTAL_AMOUNT
1	2000

Question 14

```
78
79 -----
80 ---Q14. Find the minimum Price per Unit in the dataset. Expected output: Min_Price_per_Unit
81 SELECT MIN(PRICE_PER_UNIT) AS Min_Price_Unit
82 FROM PRACTICAL1.DATASET.RETAIL_SALES;
83
84
85
86
87
88
89
```

Results (just now)

Table Chart 1 row 62ms

	# MIN_PRICE_UNIT
1	25

Question 15

my workspace / Practical.sql

```
--5. GROUP BY Statement
86
87 --Q15. Find the number of transactions per Product Category. Expected output: Product Category,
88 --- Transaction_Count
89
90 SELECT PRODUCT_CATEGORY,
91        COUNT(TOTAL_AMOUNT) AS Transaction_Count
92 FROM PRACTICAL1.DATASET.RETAIL_SALES
93 GROUP BY PRODUCT_CATEGORY;
94
95
```

Results (just now)

Table Chart 3 rows 469ms

	PRODUCT_CATEGORY	TRANSACTION_COUNT
1	Clothing	351
2	Beauty	307
3	Electronics	342

Question 16

```
93 GROUP BY PRODUCT_CATEGORY;
94
95 -----
96 ---Q16. Find the total revenue (Total Amount) per gender. Expected output: Gender, Total_Revenue
97
98 SELECT GENDER,
99        SUM(TOTAL_AMOUNT) AS Total_Revenue
100 FROM PRACTICAL1.DATASET.RETAIL_SALES
101 GROUP BY GENDER;
102
```

Results (just now)

Table Chart 2 rows 529ms

	GENDER	TOTAL_REVENUE
1	Male	223160
2	Female	232840

QUESTION 17

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```

102 -----
103 --Q17. Find the average Price per Unit per product category. Expected output: Product Category,
104      Average_Price
105
106 SELECT PRODUCT_CATEGORY,
107        AVG(PRICE_PER_UNIT) AS Average_Price
108 FROM PRACTICAL1.DATASET.RETAIL_SALES
109 GROUP BY PRODUCT_CATEGORY;

```

results (just now)

Table Chart 3 rows 87ms

	PRODUCT_CATEGORY	AVERAGE_PRICE
1	Beauty	184.055375
2	Clothing	174.287749
3	Electronics	181.900585

QUESTION 18

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```

108 -----
109 --6. HAVING Clause
110 ---Q18. Find the total revenue per product category where total revenue is greater than 10,000. Expected
111      output: Product Category, Total_Revenue
112
113 SELECT PRODUCT_CATEGORY,
114        SUM(TOTAL_AMOUNT) AS Total_Revenue
115 FROM PRACTICAL1.DATASET.RETAIL_SALES
116 GROUP BY PRODUCT_CATEGORY
117        HAVING Total_Revenue > 10000;

```

Results (just now)

Table Chart 3 rows 88ms

	PRODUCT_CATEGORY	TOTAL_REVENUE
1	Beauty	143515
2	Clothing	155580
3	Electronics	156905

QUESTION 19

My Workspace > Practical.sql

```

116
117 -----
118 --Q19. Find the average quantity per product category where the average is more than 2. Expected output:
119      Product Category, Average_Quantity
120      SELECT PRODUCT_CATEGORY,
121             AVG(QUANTITY) AS Average_Quantity
122      FROM PRACTICAL1.DATASET.RETAIL_SALES
123      GROUP BY PRODUCT_CATEGORY
124      HAVING Average_Quantity > 2;

```

Results (just now)

Table Chart 3 rows 765ms

	PRODUCT_CATEGORY	AVERAGE_QUANTITY
1	Beauty	2.511401
2	Clothing	2.547009
3	Electronics	2.482456

QUESTION 20

```

125 -----
126 -----7. CASE Statement
127 --Q20. Display a column called Spending_Level that shows 'High' if Total Amount > 1000, otherwise 'Low'.
128      Expected output: Transaction ID, Total Amount, Spending_Level
129
130      SELECT TRANSACTION_ID,
131             TOTAL_AMOUNT,
132             CASE
133               WHEN TOTAL_AMOUNT > 1000 THEN 'HIGH'
134               ELSE 'LOW'
135             END AS Spending_Level
136      FROM PRACTICAL1.DATASET.RETAIL_SALES;

```

Results (just now)

Table Chart 1,000 rows 19ms

	TRANSACTION_ID	TOTAL_AMOUNT	SPENDING_LEVEL
1	1	150	LOW
2	2	1000	LOW
3	3	30	LOW
4	4	500	LOW
5	5	100	LOW
6	6	30	LOW
7	7	50	LOW

QUESTION 21

```

136 --Q21. Display a new column called Age_Group that labels customers as:
137 --
138 --'Youth' if Age < 30.
139 --'Adult' if Age is between 30 and 59.
140 --'Senior' if Age >= 60
141 --Expected output: Customer ID, Age, Age_Group
142 SELECT CUSTOMER_ID,
143        AGE,
144        CASE
145        WHEN AGE < 30 THEN 'YOUTH'
146        WHEN AGE BETWEEN 30 AND 59 THEN 'Adult'
147        ELSE 'SENIOR'
148        END AS AGE_GROUP
149 FROM PRACTICAL1.DATASET.RETAIL_SALES;

```

Results (1 minute ago)

Download results

Table

Chart



1,000 rows



67ms



	A CUSTOMER_ID	# AGE	:	A AGE_GROUP
18	CUST018		47	Adult
19	CUST019		62	SENIOR
20	CUST020		22	YOUTH
21	CUST021		50	Adult